

Week 10 – Capstone Progress Documentation

Project Id: PW26_MG_02

Objective

The objective is to develop an **Adaptive Market Regime Detection Module** that can identify market conditions such as bullish, bearish, volatile, and sideways trends by analyzing stock price movements and selected indicators. The goal is to help the platform interpret market behavior and provide useful analytical insights for investment analysis.

Progress Update

Work has started on the Adaptive Market Regime Detection Module. The basic structure of the regime detection pipeline has been designed, and development of the feature extraction layer has begun to process market signals required for identifying market regimes. Indicators such as price momentum, moving averages, and volatility measures were selected and implemented for early testing. Initial rule-based experiments were conducted to observe how these indicators respond during different market conditions and to understand how they may help identify regime transitions.

Other platform tasks were also implemented. Symbol lookup functionality was developed to retrieve stock symbols. Watchlist management features were created to allow users to track selected stocks. Stock catalog and listing APIs were implemented to manage available stock information within the platform.

Next Steps

The next phase will focus on improving the Adaptive Market Regime Detection Module. The regime detection logic will be refined to better interpret indicator signals and improve classification of market conditions. Additional indicators will be evaluated and integrated to strengthen the detection process. Testing will be performed using historical market data to study how the indicators behave across different time periods and market environments.

Further work will continue on strengthening the platform components. Improvements will be made to stock data services, portfolio tracking functions, and transaction logging mechanisms to ensure reliable data management and consistent system performance across the platform.

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